

Analysis group meeting

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09/22/25



MISSISSIPPI STATE
UNIVERSITY™

Pass1 status



Total Jobs

177,186



Completed

51,988



Running

10,006



Pending

114,690



Problems

0

Overall Progress



Success Rate
29.3%

Active Workflows
3

Complete Workflows
4

Workflows Overview

Workflow Name	Status	Jobs	Progress	Completion	Problems
rgd-dmra-pass1_empty_zero_field-...	✔ Complete	239/239		100%	-
rgd-dmra-pass1_empty_nominal-183...	✔ Complete	113/113		100%	-
rgd-dmra-pass1_LD2_2-18305x144	⌚ Waiting	0/40,076		0%	-
rgd-dmra-pass1_LD2_1-18431x16	✔ Complete	7,702/7,702		100%	-
rgd-dmra-pass1_CxC_2-18339x116	⚙ Running	16,150/36,705		44%	-
rgd-dmra-pass1_CxC_1-18445x13	✔ Complete	6,873/6,873		100%	-
rgd-dmra-pass1_CuSn_2-18347x294	⚙ Running	6,371/70,791		9%	-
rgd-dmra-pass1_CuSn_1-18561x33	⚙ Running	14,540/14,687		99%	-

Pass1 status

3

Skim for CuSn Pass1: /volatile/clas12/ouillon/skim_pass1_RGD/CuSn/

Electron Identification

4

□ Electron Refinements Cuts:

- SF vs p: sector, target, and torus polarity dependent
- DC Fiducial cuts: R1: $0.07 \theta_{DC} + 2.66 \text{ cm}$ R2: $0.02 \theta_{DC} + 2.80 \text{ cm}$ R3: $0.30 \theta_{DC} + 3.27 \text{ cm}$
- ECal Fiducial cut: V and W > 9 cm
- Momentum: $p > 0.8 \text{ GeV}$ (The electron reconstruction can't be trusted below 0.8 GeV)
- Vz vertex: sector, target, and torus polarity dependent

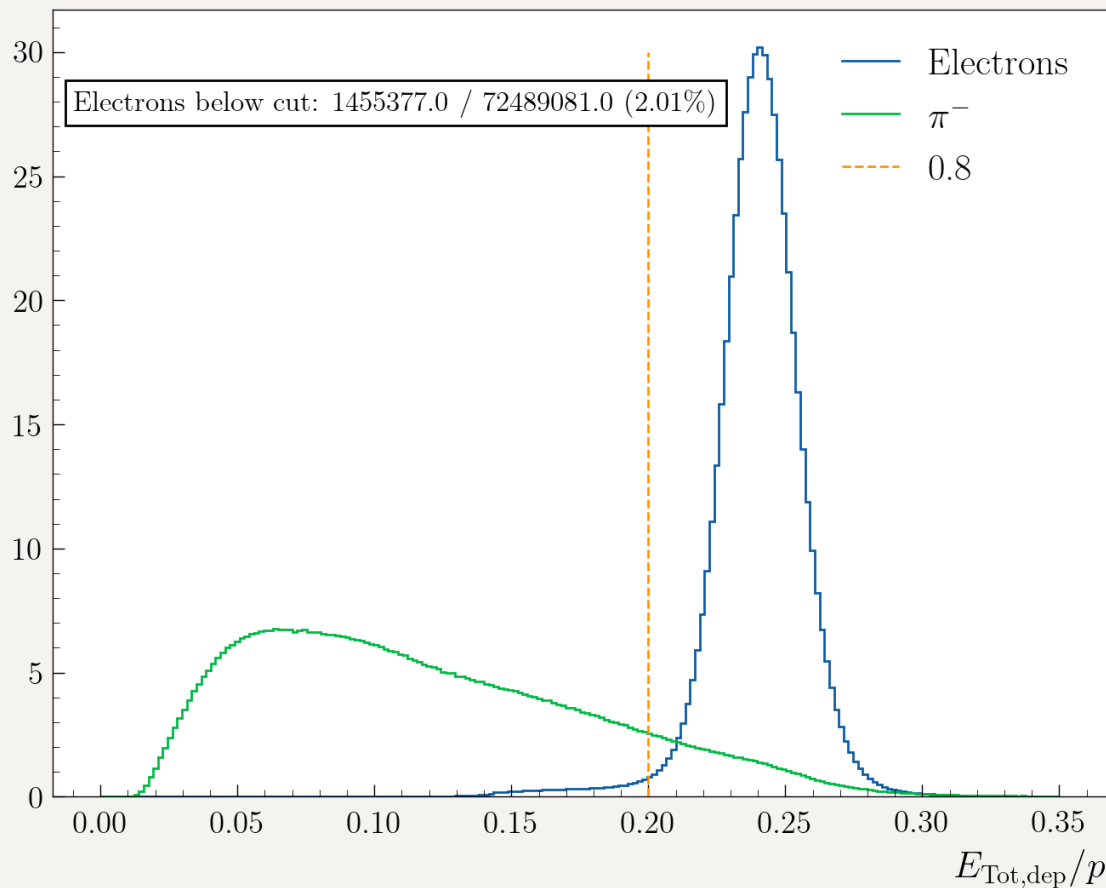
Target: LD2

Total events:	159223181	Now the main cut is the DC fiducial: Reduces total electron count by 20%, which is too much
With electron:	111103379 (100.00%)	
SF:	107937848 (97.15%)	
+edges:	88379993 (79.55%)	
+v/w:	88379947 (79.55%)	
+p>0.8 GeV:	88377712 (79.55%)	
+vz cut:	84099766 (75.70%)	

Electron Identification

5

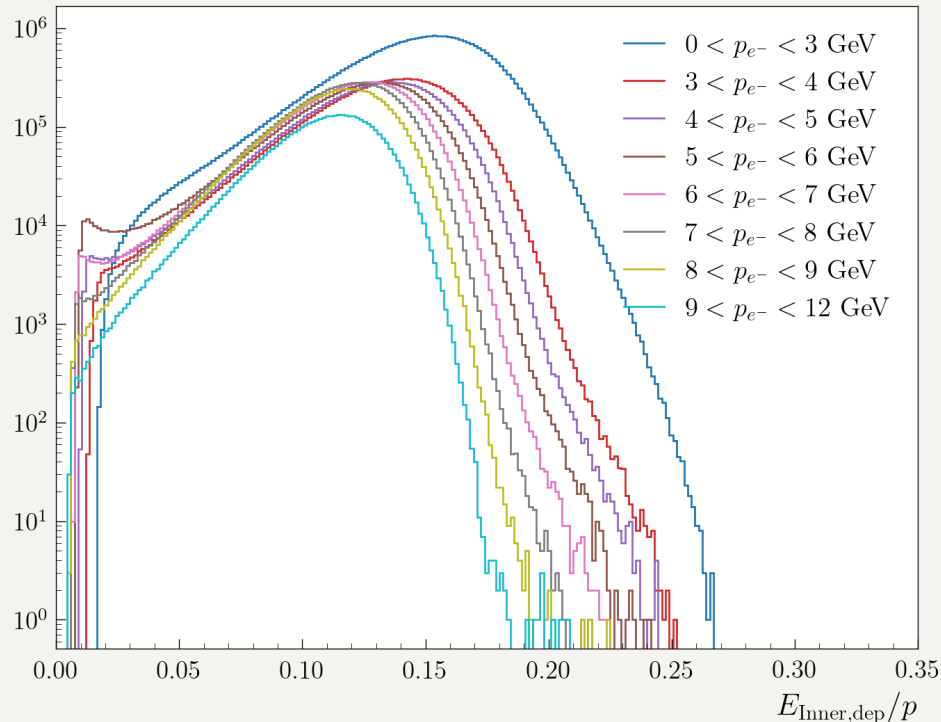
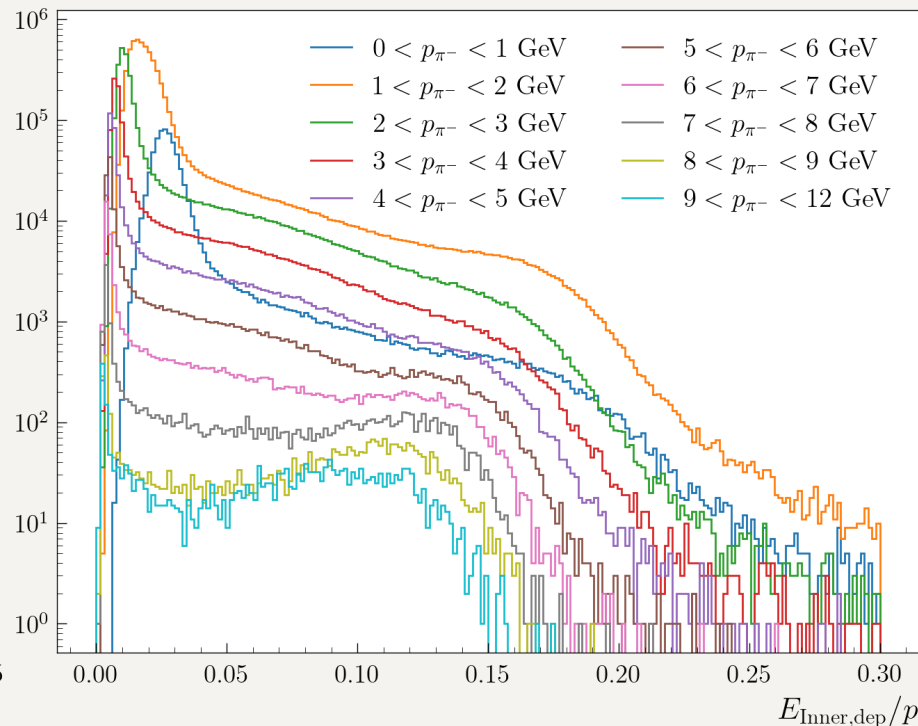
- Comparison between SF for electrons and π^-
- Electron identify with pid=11 and status
- Pi- identify with pid=-211



Electron Identification

6

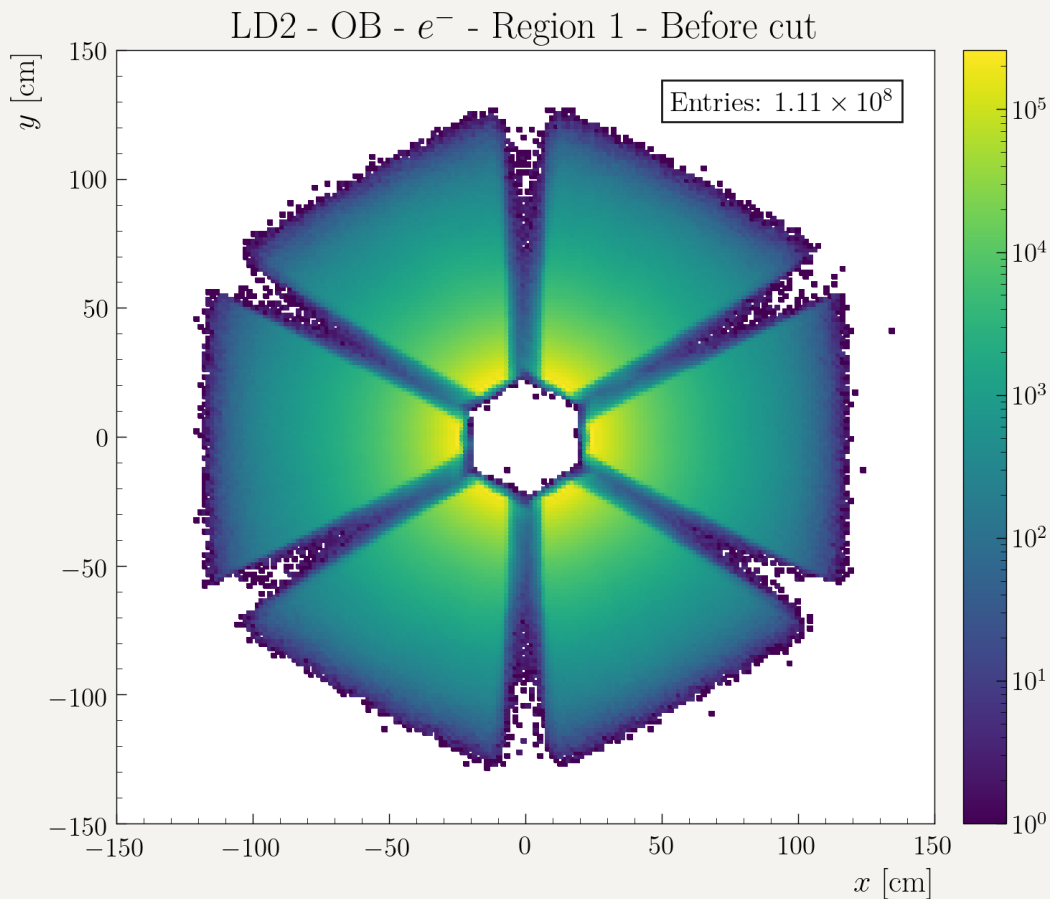
Comparison between SF in the Inner part for electrons and π^-

LD2 - OB - e^- LD2 - OB - π^- 

Electron Identification

7

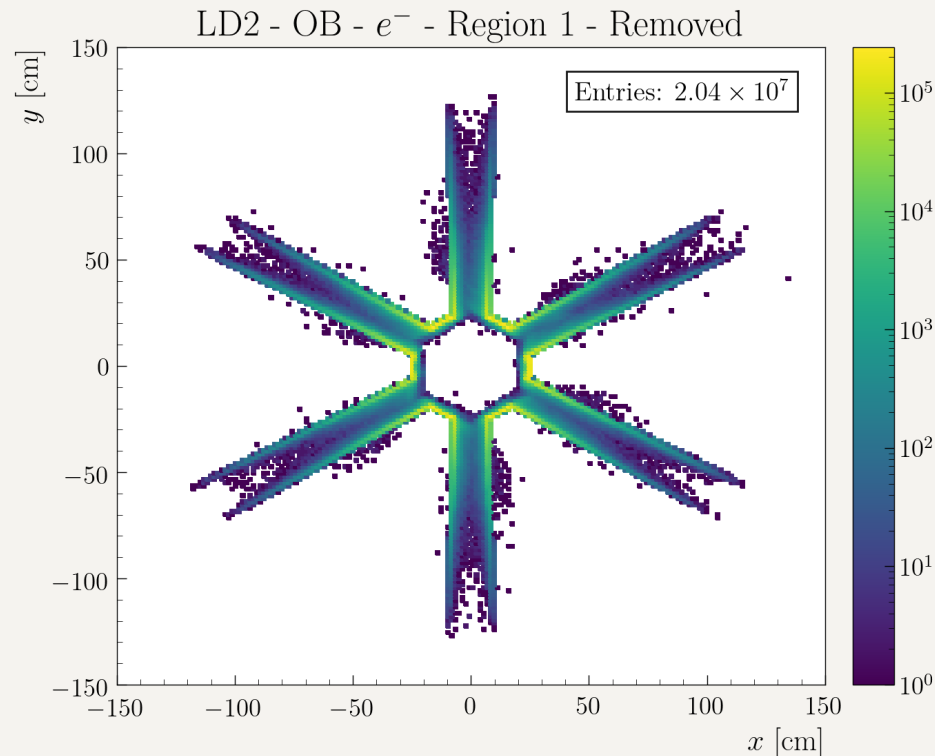
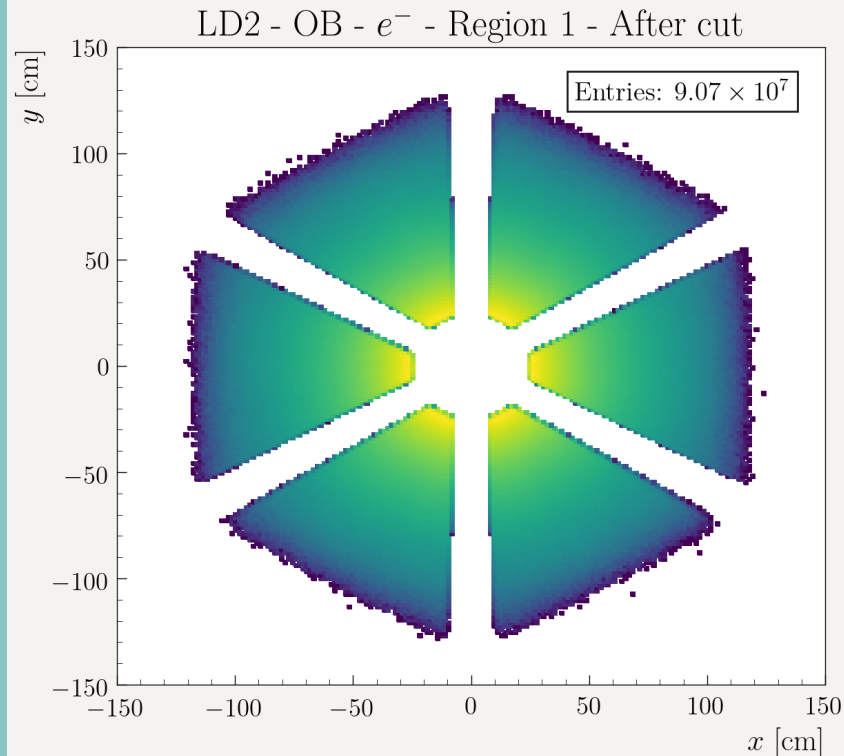
- Cut on the DC edge variation as a function of the local θ in the DC plan.
- One cut for each DC region:
 - Region 1: $0.07 \theta_{\text{DC}} + 2.66 \text{ cm}$
 - Region 2: $0.02 \theta_{\text{DC}} + 2.80 \text{ cm}$
 - Region 3: $0.30 \theta_{\text{DC}} + 3.27 \text{ cm}$



Electron Identification

8

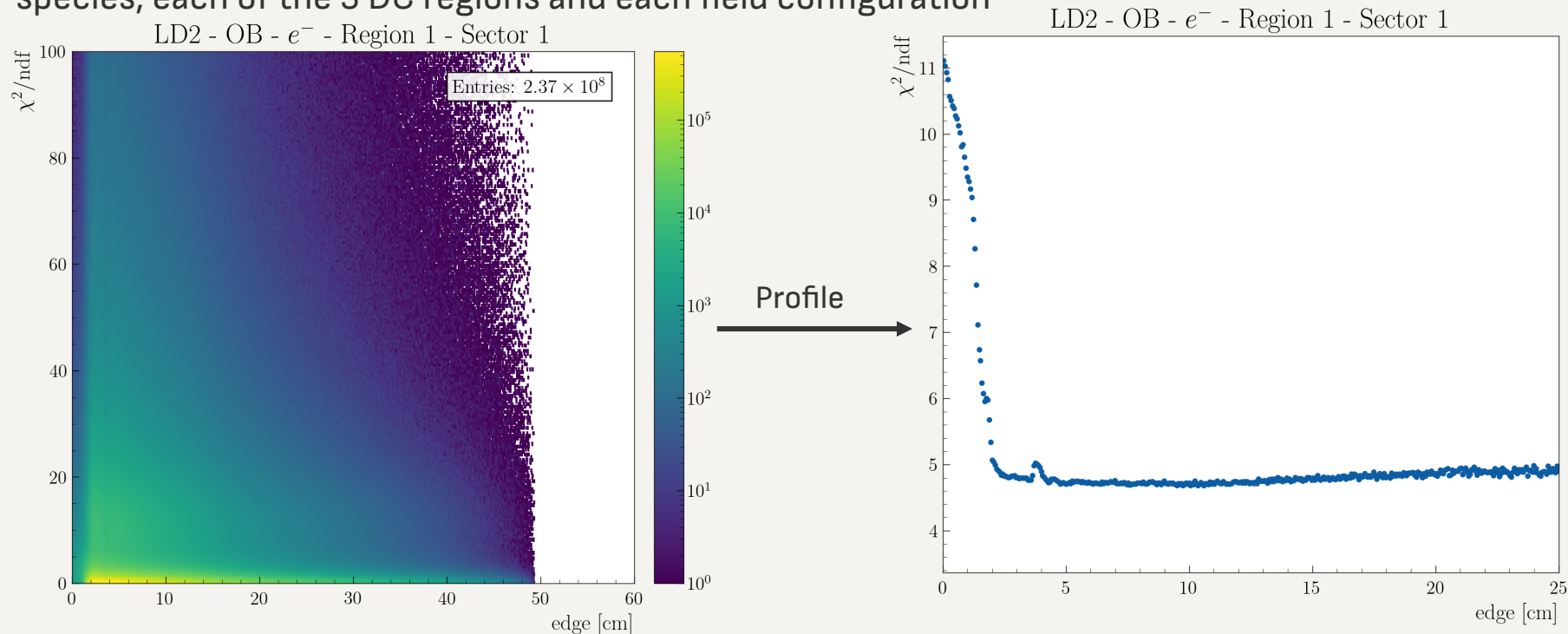
- Cut on the DC edge variation as a function of the local θ in the DC plan.
- I think I cut to much, specially at low θ , maybe use 1 or 2σ cut instead of 3



Electron Identification

9

- In the RG-A/M analysis note, the fiducial cuts for the DC are based on the distribution of χ^2/NDF from the track as a function of the edge variables adjusted individually for each particle species, each of the 3 DC regions and each field configuration

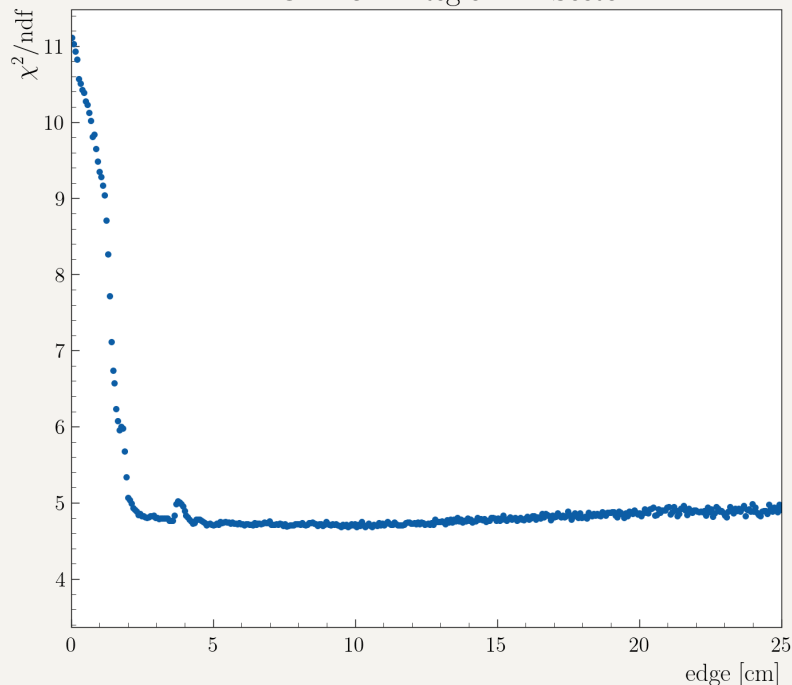


Electron Identification

10

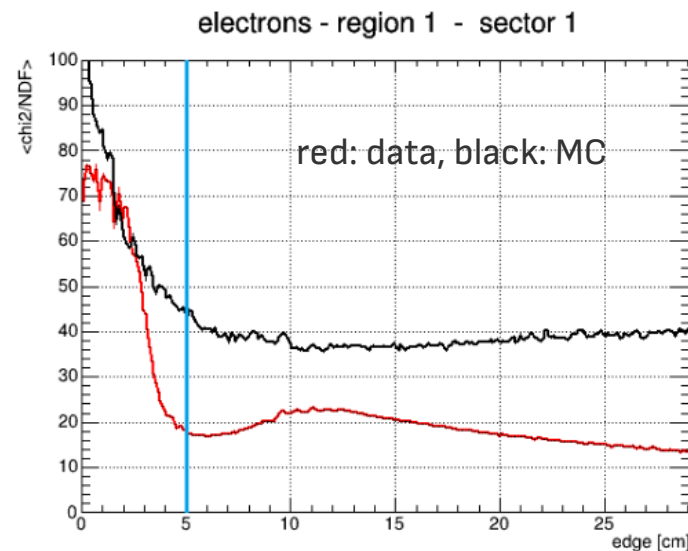
- In the RG-A/M analysis note, the fiducial cuts for the DC are based on the distribution of χ^2/NDF from the track as a function of the edge variables adjusted individually for each particle species, each of the 3 DC regions and each field configuration

LD2 - OB - e^- - Region 1 - Sector 1



The goal is to find where the distribution starts to rise
Here, around 2cm.

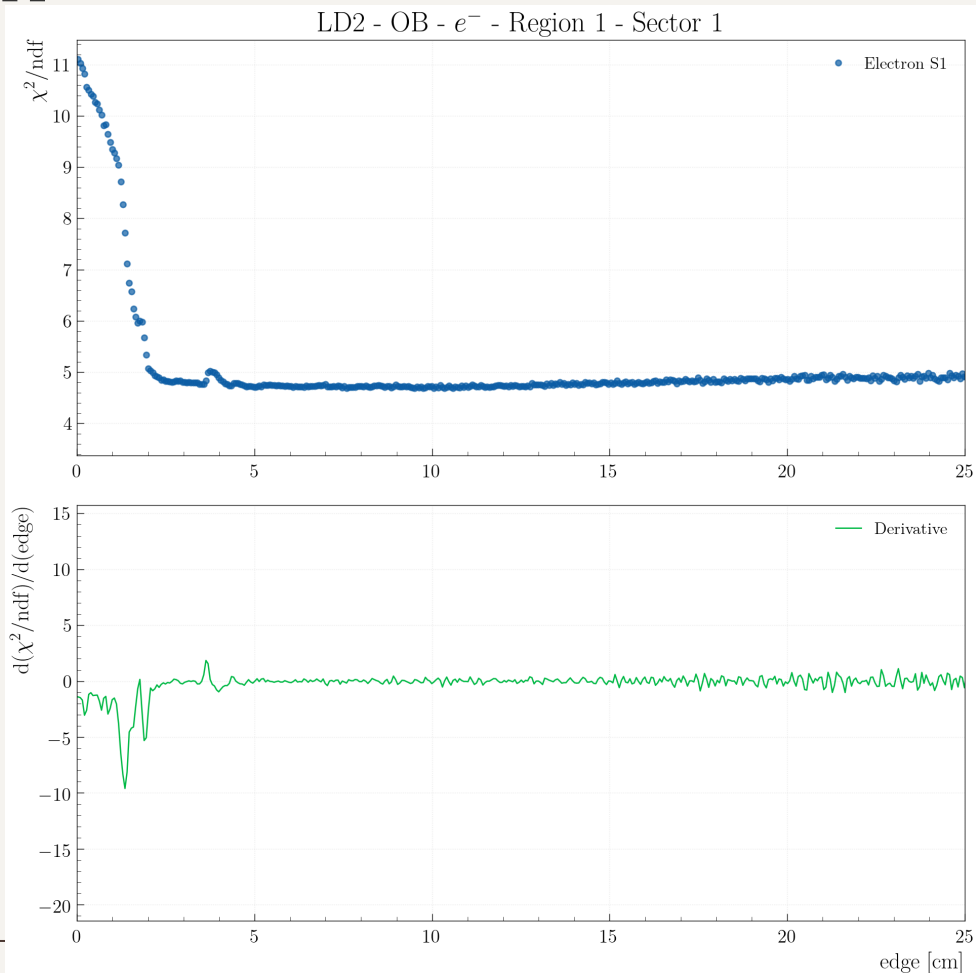
RG-A and M seem to choose the values for the cuts
by eye.



Electron Identification

11

- Using the derivative is a bit difficult here: Many random fluctuations.

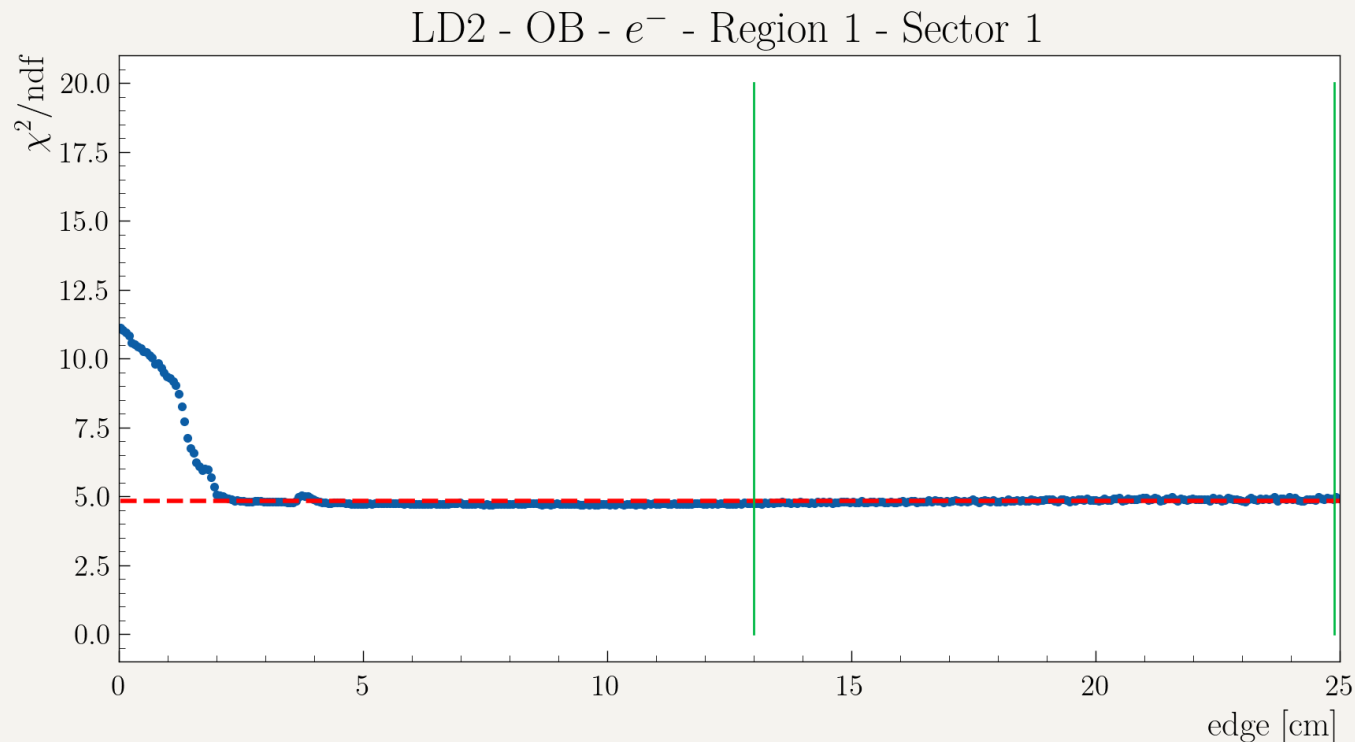


Electron Identification

12

□ My solution:

- Fit the region between 13 and 25 cm with a constant.



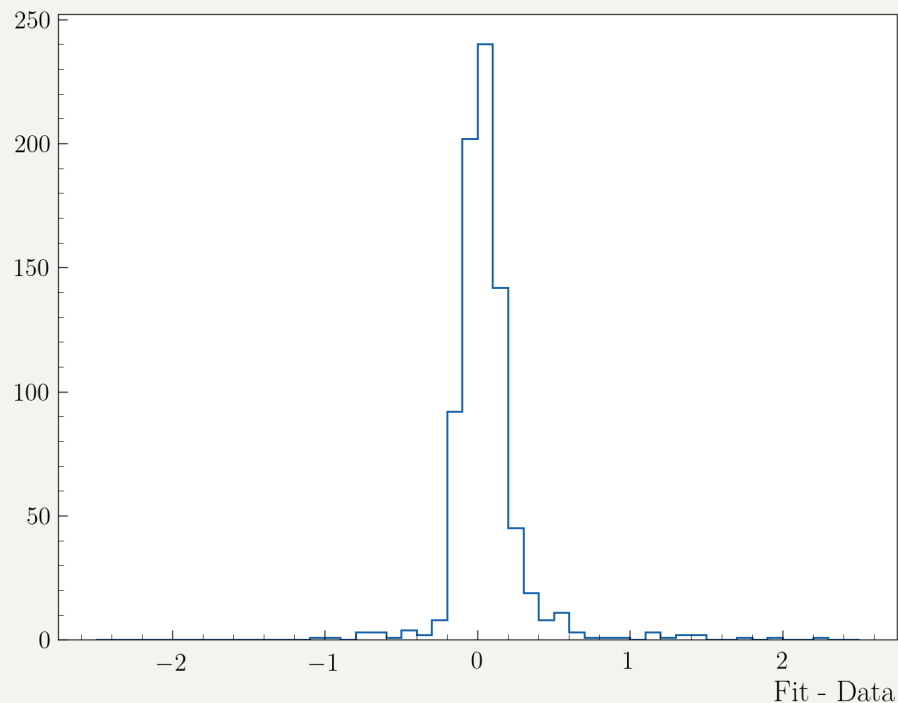
Electron Identification

13

□ My solution:

- Fit the region between 13 and 25 cm with a constant.
- Compute the distance between the fit and the data.
- Compute the standard deviation of the difference:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}, \quad \text{where} \quad \mu \equiv \frac{1}{N} \sum_{i=1}^N x_i.$$



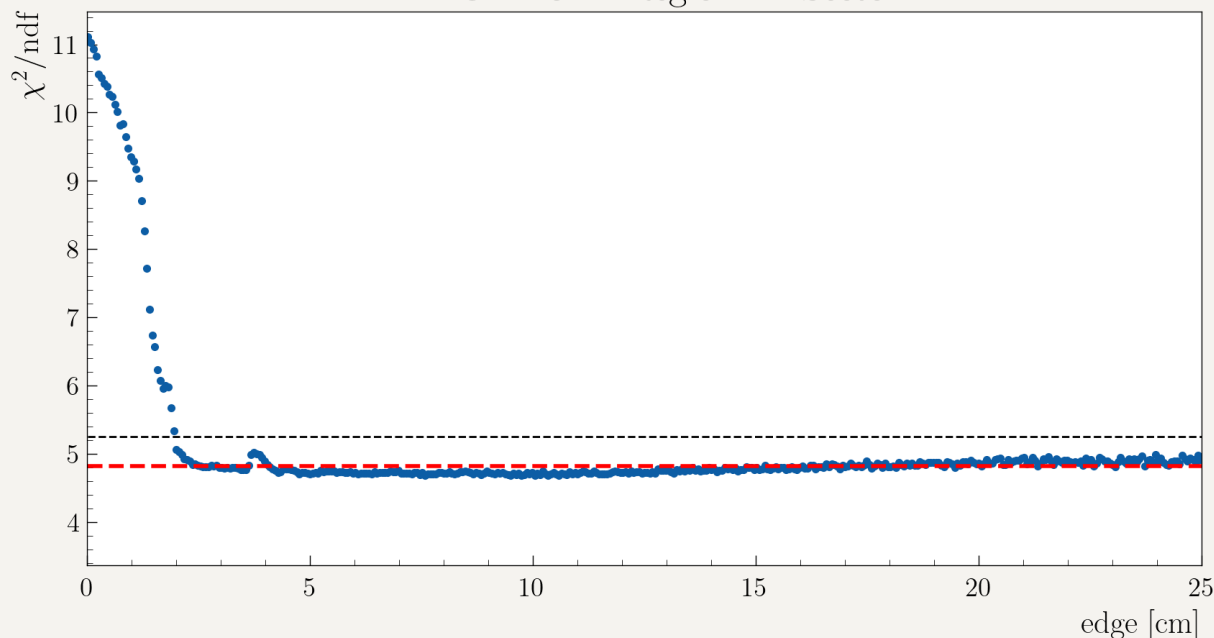
Electron Identification

14

□ My solution:

- Fit the region between 13 and 25 cm with a constant.
- Compute the distance between the fit and the data.
- Compute the standard deviation of the difference and plot fit + 0.5std

LD2 - OB - e^- - Region 1 - Sector 1

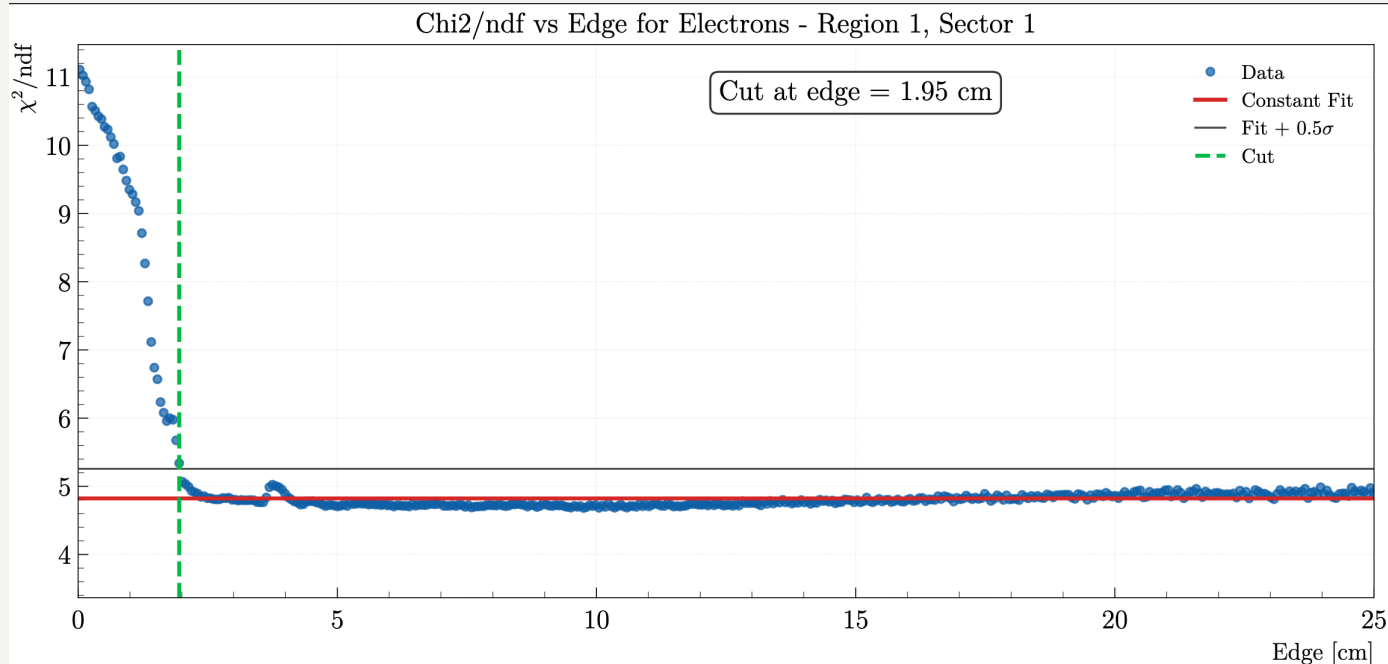


Electron Identification

15

□ My solution:

- Fit the region between 13 and 25 cm with a constant.
- Compute the distance between the fit and the data.
- Compute the standard deviation of the difference and plot fit + 0.5std
- Find the rightmost sequence of 5 consecutive points exceeding fit + 0.5std and the end of the sequence is the cut



Electron Identification

□ My solution:

- Fit the region between 13 and 25 cm with a constant.
- Compute the distance between the fit and the data.
- Compute the standard deviation of the difference and plot fit + 0.5std
- Find the rightmost sequence of 5 consecutive points exceeding fit + 0.5std and the end of the sequence is the cut

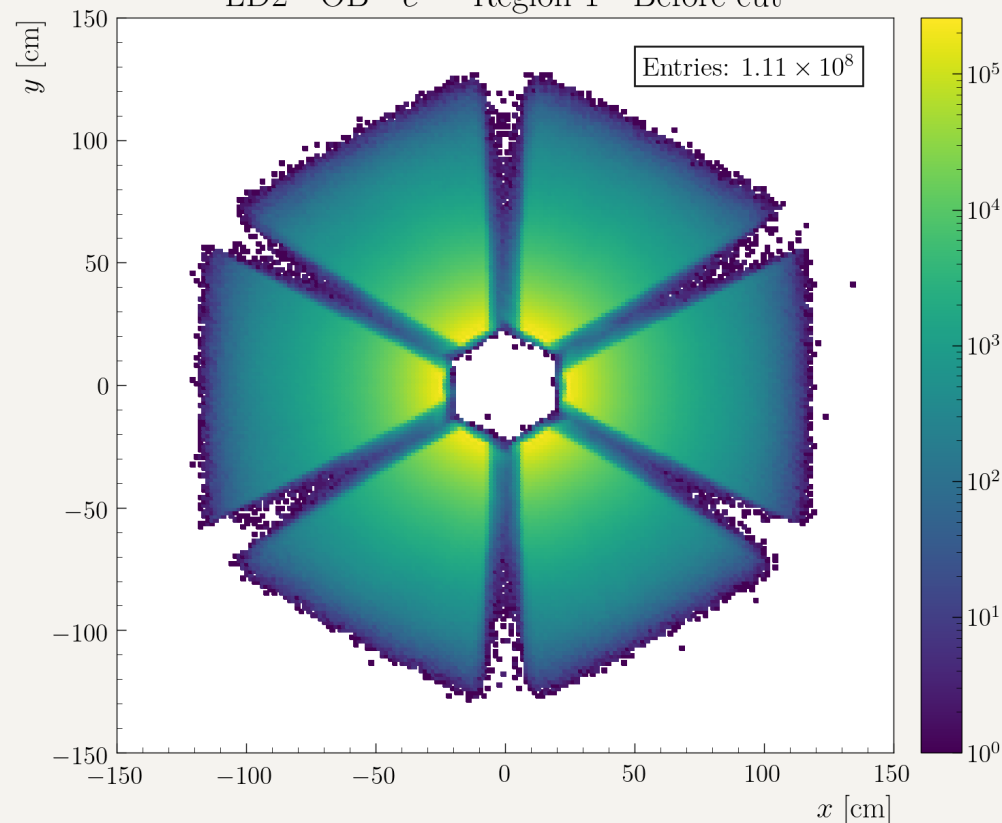
```
=====
edge_vs_chi2ndf_region1_s1_e.root | Cut at 1.95 cm
edge_vs_chi2ndf_region1_s2_e.root | Cut at 1.71 cm
edge_vs_chi2ndf_region1_s3_e.root | Cut at 1.29 cm
edge_vs_chi2ndf_region1_s4_e.root | Cut at 1.47 cm
edge_vs_chi2ndf_region1_s5_e.root | Cut at 1.89 cm
edge_vs_chi2ndf_region1_s6_e.root | Cut at 1.77 cm
edge_vs_chi2ndf_region2_s1_e.root | Cut at 1.95 cm
edge_vs_chi2ndf_region2_s2_e.root | Cut at 2.07 cm
edge_vs_chi2ndf_region2_s3_e.root | Cut at 1.89 cm
edge_vs_chi2ndf_region2_s4_e.root | Cut at 2.37 cm
edge_vs_chi2ndf_region2_s5_e.root | Cut at 1.77 cm
edge_vs_chi2ndf_region2_s6_e.root | Cut at 1.95 cm
edge_vs_chi2ndf_region3_s1_e.root | Cut at 8.07 cm
edge_vs_chi2ndf_region3_s2_e.root | Cut at 8.85 cm
edge_vs_chi2ndf_region3_s3_e.root | Cut at 8.97 cm
edge_vs_chi2ndf_region3_s4_e.root | Cut at 8.31 cm
edge_vs_chi2ndf_region3_s5_e.root | Cut at 9.09 cm
edge_vs_chi2ndf_region3_s6_e.root | Cut at 9.15 cm
```


Electron Identification

17

Effect on y vs x distribution:

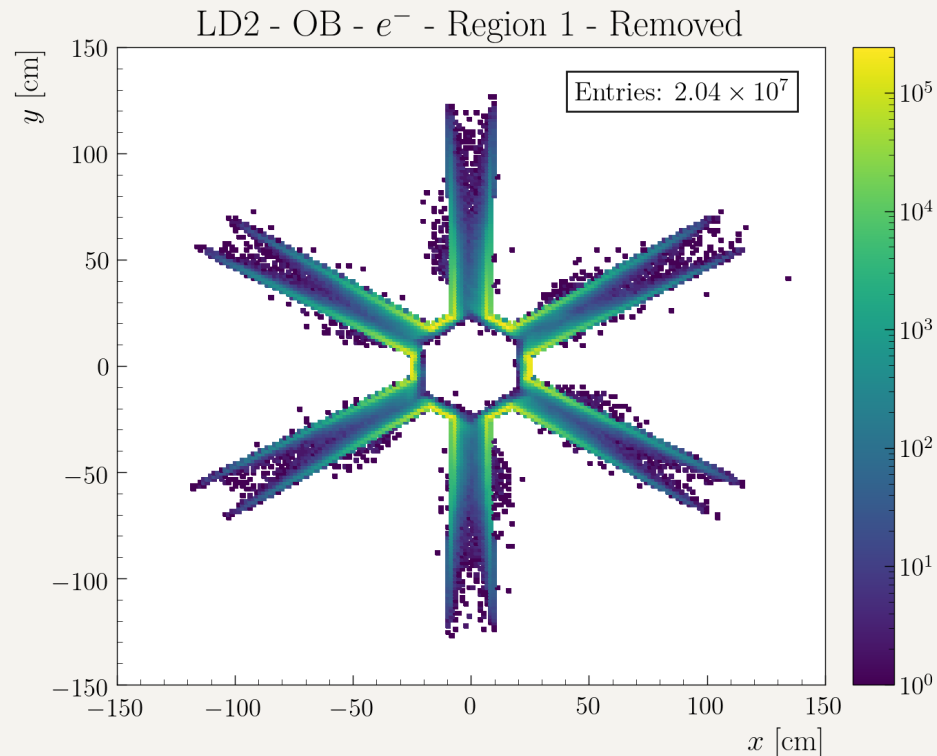
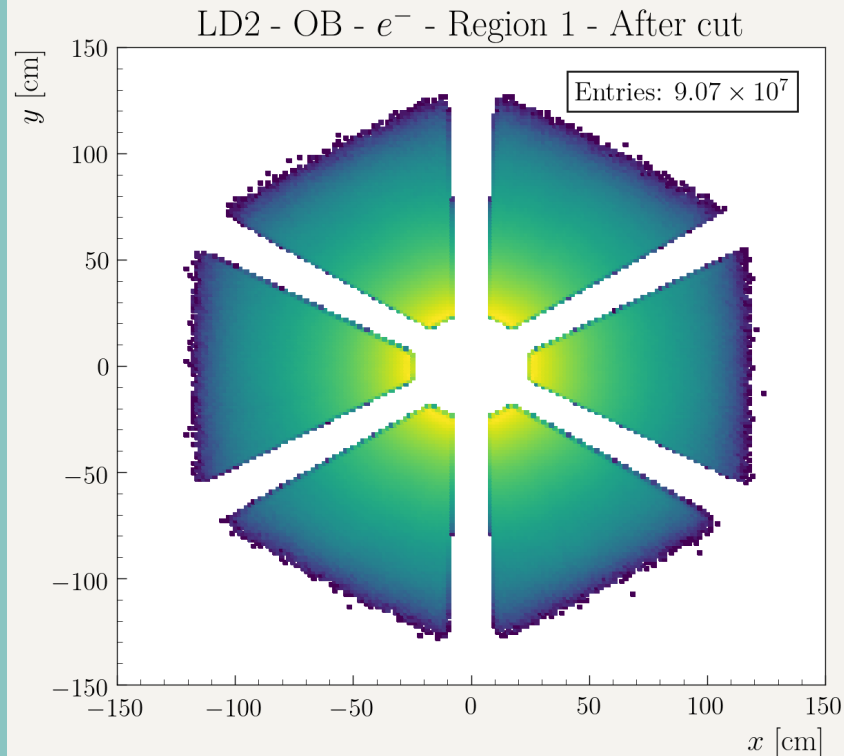
LD2 - OB - e^- - Region 1 - Before cut



Electron Identification

18

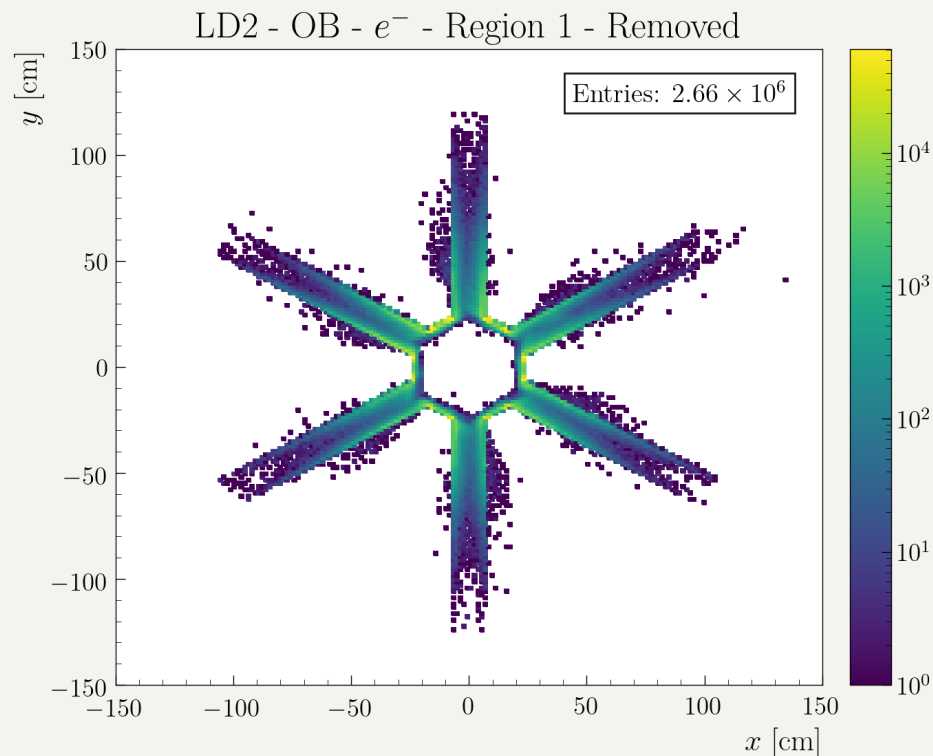
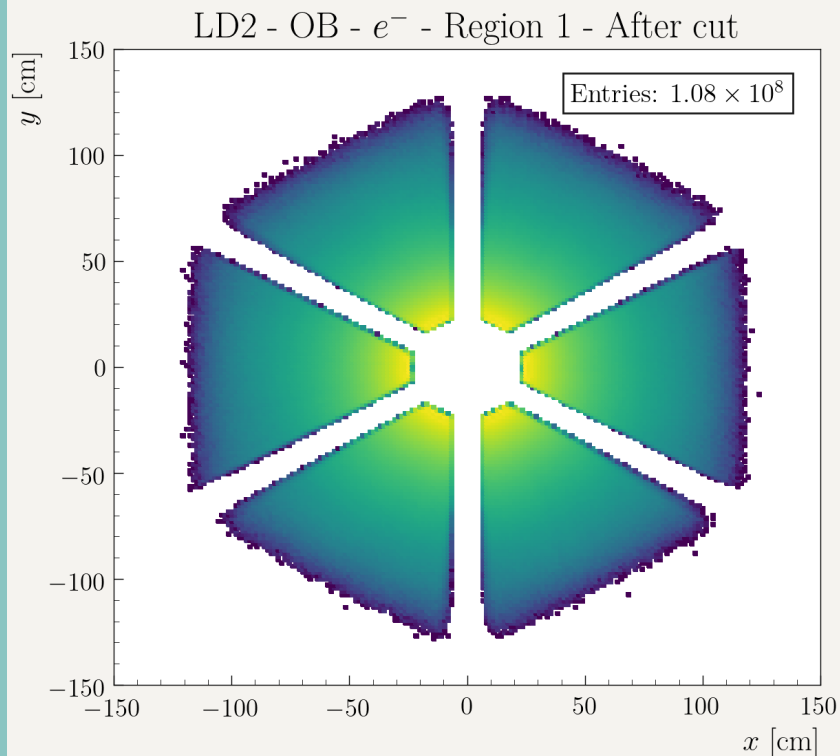
- Cut on the DC edge variation as a function of the local θ in the DC plan.
- I think I cut to much, specially at low θ , maybe use 1 or 2σ cut instead of 3



Electron Identification

19

Effect on y vs x distribution: Survival fraction: 97.60 %



CLAS Collaboration meeting

20

□ Plan for the presentation:

- Physics and RG slides
- Cooking Status and progress
- Electron Identification (SF, DC fiducials, ECAL, Vz, (maybe SFPCAL vs SFInner?))
- π^0 analysis:
 - background subtraction
 - mass of the 2 photons for each target
 - multidimensional multiplicity ratio
 - $\cos \phi$ modulation
- Daniel/Suman presentation 1 slide per side